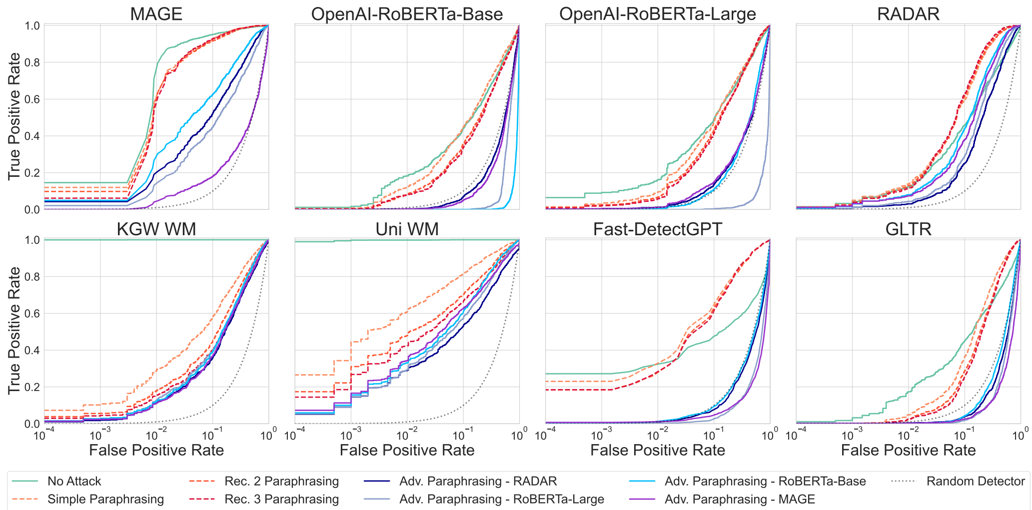

Figure 3: **ROC curves illustrating the AI text detection performance on several deployed detectors, including neural network-based, watermark-based, and zero-shot detectors.** The false positive rate (FPR) axis is displayed in log-scale to highlight fine-grained distinctions in the low-FPR region. It can be observed that adversarial paraphrasing consistently and significantly reduces the detection performance across all deployed detectors when compared to the baselines.

	RoBERTa-Large		**RoBERTa-Base**		**MAGE**		**RADAR**		
	AUC (↓)	**T@1%F (↓)**	**AUC (↓)**	**T@1%F (↓)**	**AUC (↓)**	**T@1%F (↓)**	**AUC (↓)**	**T@1%F (↓)**	**Rating**
No Attack	0.789	0.163	0.745	0.182	0.975	0.768	0.767	0.124	–
Simple Paraphrase	0.794	0.096	0.762	0.119	0.970	0.616	0.881	0.140	4.75 ± 0.54
Rec. Para. 2	0.777	0.069	0.712	0.082	0.967	0.609	0.885	0.130	4.47 ± 0.67
Rec. Para. 3	0.779	0.059	0.706	0.079	0.969	0.585	0.893	0.117	4.26 ± 0.74
AdvPara (RADAR)	0.538	0.013	0.464	0.004	0.815	0.201	0.723	0.031	4.45 ± 0.79
AdvPara (RoBERTa-Large)	0.147	0.000	0.323	0.000	0.769	0.142	0.768	0.044	4.48 ± 0.77
AdvPara (RoBERTa-Base)	0.557	0.006	0.110	0.000	0.861	0.291	0.826	0.080	4.54 ± 0.59
AdvPara (MAGE)	0.543	0.011	0.435	0.003	0.518	0.045	0.807	0.074	4.54 ± 0.70
	KGW WM		**Uni WM**		**Fast-DetectGPT**		**GLTR**		
	AUC (↓)	**T@1%F (↓)**	**AUC (↓)**	**T@1%F (↓)**	**AUC (↓)**	**T@1%F (↓)**	**AUC (↓)**	**T@1%F (↓)**	**Rating**
No Attack	1.000	1.000	1.000	0.999	0.666	0.323	0.726	0.174	–
Simple Paraphrase	0.841	0.295	0.927	0.609	0.873	0.326	0.782	0.049	4.75 ± 0.54
Rec. Para. 2	0.790	0.181	0.881	0.480	0.867	0.275	0.745	0.026	4.47 ± 0.67
Rec. Para. 3	0.762	0.155	0.858	0.424	0.867	0.276	0.739	0.025	4.26 ± 0.74
AdvPara (RADAR)	0.741	0.117	0.777	0.291	0.452	0.009	0.433	0.004	4.45 ± 0.79
AdvPara (RoBERTa-Large)	0.769	0.131	0.827	0.294	0.338	0.003	0.400	0.001	4.48 ± 0.77
AdvPara (RoBETa-Base)	0.769	0.125	0.852	0.332	0.480	0.012	0.481	0.001	4.54 ± 0.59
AdvPara (MAGE)	0.750	0.113	0.831	0.344	0.301	0.011	0.338	0.003	4.54 ± 0.70

Table 2: **Detection performance of eight different deployed detectors in distinguishing between AI-generated and human-written text under different attack scenarios.** Metrics reported include AUC and TPR at 1% FPR for each detector. Additionally, we present the mean ± standard deviation of quality ratings given by GPT-4o. Further details on text quality analysis are provided in Section 5.

based and zero-shot detectors. The superior performance of our attack highlights the importance of the guidance signal provided by a trained detector, which effectively steers the paraphrasing to adversarially align with the statistical characteristics of human-authored content.

Universality. We find that adversarial paraphrasing guided by one detector can reduce the detection rates of all the other detectors we consider, showing the universal transferability of our method. We also find that any target deployed detector can be evaded by adversarial paraphrasing guided by any trained detectors we consider in our study. We present these findings by plotting the complete transferability matrix in Figure 4, presenting the relative drop in T@1%F for all guidance–deployment combinations of detectors.

Our results show that adversarial paraphrasing is robust to the choice of guidance detectors that we consider in our study. On average, we observe a relative drop in T@1%F of 84.94% when using MAGE as guidance, 86.89% with RADAR, 80.75% with OpenAI-RoBERTa-Base, and 87.88% with OpenAI-RoBERTa-Large. Although different guidance detectors may yield slightly varying degrees of transferability depending on the deployed detector, all our adversarially paraphrased outputs, agnostic

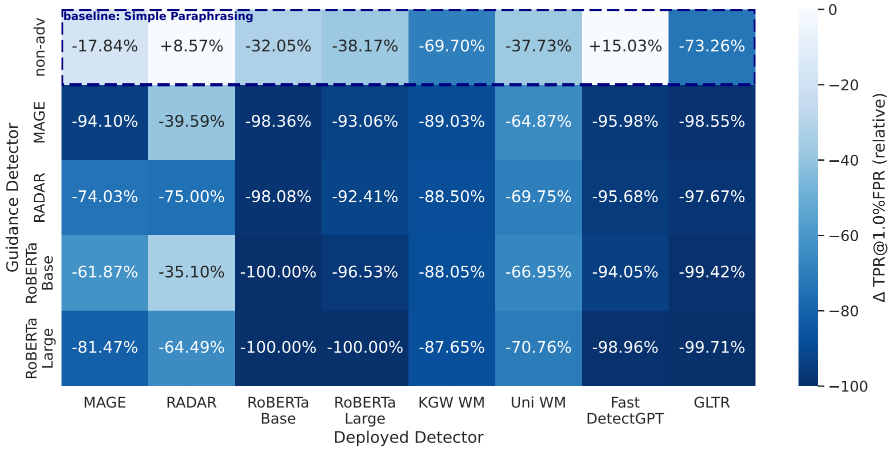

Figure 4: **Relative drop in T@1%F across all combinations of guidance and deployed detectors. The first row corresponds to simple (non-adversarial) paraphrasing baseline [15].** On average, simple paraphrasing leads to a 30.27% relative drop in T@1%F. In comparison, adversarial paraphrasing achieves significantly higher reductions—84.94% with MAGE as guidance, 86.89% with RADAR, 80.75% with OpenAI-RoBERTa-Base, and 87.88% with OpenAI-RoBERTa-Large. These results highlight both the universal effectiveness and transferability of our attack.

of the guidance detector, consistently lead to significant reductions in T@1%F compared to simple paraphrasing. This further underscores the universal effectiveness and transferability of our attack.

4.3 Efficiency of Adversarial Paraphrasing

Compared to simple paraphrasing, our method requires running a surrogate AI text detector at each decoding step by design. This naturally raises the question of how much latency this process introduces and whether it affects efficiency.

To evaluate this, we conducted adversarial paraphrasing on 100 randomly selected text samples, with five trials per configuration, using all four guidance detectors. To simplify measurement and report time on a per-sample basis (rather than per batch), we used a batch size of 1 during these trials. In practical settings, including the main experiments described in Section 4.2, we employ larger batch sizes, which substantially reduce total runtime. The average paraphrasing time per sample across five trials is reported as the mean ± standard deviation (in seconds[3]) in Table 3. As shown, most guidance detectors introduce only minor latency compared to simple paraphrasing. The higher latency observed with MAGE arises from its LongFormer-based architecture, which requires longer inference time than the RoBERTa-based models.

Method	Run time
Simple Paraphrase	7.18 ± 0.13
AdvPara (roblarge)	10.20 ± 0.18
AdvPara (robbase)	8.64 ± 0.11
AdvPara (mage)	16.71 ± 0.74
AdvPara (radar)	9.69 ± 0.20

Table 3: Per-sample run time (in seconds) of adversarial paraphrasing over 100 randomly selected samples.

Overall, latency is primarily determined by the computational complexity of the guidance detector. From a computational cost perspective, the detector adds minimal overhead in terms of FLOPs relative to the paraphrasing LLM, which dominates total computation. In our experiments, the paraphrasing LLM contains approximately 8 billion parameters, whereas the detectors range from 100 to 350 million parameters—less than 5% of the paraphraser's size for the larger detectors and under 2% for the smaller ones.

4.4 Ablation Studies

Other than the guidance detector choice, the two main hyper-parameters related to our work are the p and k used in top-p and top-k masking. In the above, we have used $p = 0.99$ and $k = 50$, where $k = 50$ is also the default k value set in HuggingFace Transformer library. To show how their settings

[3] Note that runtime per sample may vary with sequence length. To ensure a fair comparison—i.e., that AdvPara generates a similar number of tokens as simple paraphrasing—we report the mean and standard deviation of token counts in Appendix B.

affect our attack, we use OpenAI-RoBERTa-Large as the guidance detector and run ablation studies on the value of p and k on 500 randomly selected texts. Defaulting to $k = 50$ and $p = 0.99$ when varying the other, we report the **mean** AUC and TPR@1%FPR across all 8 deploy detectors in Table 4.

Method	mean AUC	mean T@1%R	Rating
No Attack	0.8419	0.4935	–
Simple Paraphrase	0.8588	0.2885	4.75 ± 0.54
AdvPara (p=0.5)	0.8606	0.2985	4.86 ± 0.45
AdvPara (p=0.8)	0.7895	0.2230	4.79 ± 0.48
AdvPara (p=0.9)	0.7309	0.1820	4.79 ± 0.48
AdvPara (p=0.95)	0.6816	0.1480	4.81 ± 0.44
AdvPara (p=0.99)	0.5433	0.0690	4.50 ± 0.67
AdvPara (p=0.991)	0.5381	0.0592	4.47 ± 0.73
AdvPara (p=0.992)	0.5244	0.0565	4.37 ± 0.76
AdvPara (k=10)	0.5596	0.0757	4.47 ± 0.77
AdvPara (k=25)	0.5448	0.0658	4.48 ± 0.73
AdvPara (k=50)	0.5433	0.0690	4.50 ± 0.67
AdvPara (k=75)	0.5421	0.0698	4.52 ± 0.73
AdvPara (k=100)	0.5426	0.0698	4.51 ± 0.70

Table 4: Attack effectiveness under different values of p and k used in top-p and top-k masking.

Fixing k at 50, we can observe that when p exceeds 0.99, the marginal gains in attack effectiveness diminish relative to the losses in text quality. Additionally, as one can imagine, further increasing slows down generation speed. Therefore, we select $p = 0.99$ as a balanced trade-off among attack effectiveness, text quality, and generation speed. With p fixed at 0.99, the values of k show relatively smaller influence, and mostly lead to comparable results. Hence we just stick with the default value in the HuggingFace Transformer library.

5 Quality Evaluation of Paraphrased Texts

We conduct a comprehensive evaluation to investigate the impact of adversarial paraphrasing on the perceived quality of AI-generated text, focusing on both semantic equivalence to the original text and clarity, fluency, and naturalness of the text itself. For this, we randomly sample 100 texts each from our three datasets (MAGE, KGW watermarked MAGE, and Unigram watermarked MAGE) and analyze them with four complementary studies: (1) perplexity scores (PPL), (2) SBERT [25] semantic similarity, (3) auto-rated quality from GPT-4o comparing paraphrased texts to their original AI counterparts, and (4) head-to-head win rates, also assessed by GPT-4o, comparing adversarial paraphrasing against simple paraphrasing. We provide representative examples of paraphrases in Table 1, with a much broader set of examples included in Appendix F to support qualitative manual inspection. Our findings highlight a nuanced trade-off between evading detectors and preserving textual quality.

Text	**PPL** (mean$\pm$std)
Original AI	14.94 ± 10.40
Original Human	15.02 ± 7.71
Simple Paraphrase	9.28 ± 3.86
AdvPara (roblarge)	14.26 ± 4.97
AdvPara (robbase)	14.86 ± 6.32
AdvPara (mage)	17.11 ± 7.33
AdvPara (radar)	14.26 ± 5.13

Table 5: Perplexity (PPL) scores for original AI-generated and human-written texts from the MAGE dataset, along with simple and adversarial paraphrased versions of the AI-generated texts.

Perplexity Analysis. We assess perplexity using LLaMA-3.1–8B-Instruct [21], comparing the original AI-generated text, simple paraphrases, and adversarial paraphrases. The results are summarized in Table 5. As shown in the table, human-written text typically exhibits higher perplexity than AI text, as human language tends to deviate more from the statistical regularities learned by LLMs. After applying simple paraphrasing, we observe a substantial improvement in the perplexity of AI text. This may be attributed to the fact that the model used for paraphrasing (LLaMA-3.1) is superior to the LLMs used for generating AI texts in the MAGE dataset (*e.g.* LLaMA). In contrast, adversarial paraphrasing yields perplexity scores that are comparable to the human texts from MAGE, which is reasonable given that our objective is to humanize AI texts.

Method	SBERT Cos. Sim.
Simple Paraphrase	0.8601 ± 0.0880
AdvPara (roblarge)	0.8082 ± 0.1006
AdvPara (robbase)	0.8128 ± 0.0985
AdvPara (mage)	0.8159 ± 0.0982
AdvPara (radar)	0.8095 ± 0.1025

Table 6: Cosine similarity of SBERT embeddings before and after paraphrasing.

SBERT Similarity. A common approach to assessing the semantic equivalence between two texts is to compute the cosine similarity between their SBERT embeddings [25]. Accordingly, we measure the cosine similarity of SBERT embeddings before and after paraphrasing, and compare adversarial paraphrasing with simple paraphrasing. The results are summarized in Table 6. Overall, although there is a slight reduction in the mean cosine similarity for adversarial paraphrasing, the values remain within an acceptable range given the high variance observed across samples.